

Supplementary Material: History-Aware Control Barrier Functions with Nonsingular Fading-Memory Kernels

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1 Purpose and scope

This companion collects the extended numerical material referenced by the six-page manuscript. Its role is to document the broader validation evidence and to show that the same history-conditioned authority mechanism persists beyond the one-state Caputo–Fabrizio (CF) realization used in the main paper.

The supplementary content is intentionally limited to three items: (i) the complete high-demand stress-test atlas, (ii) the bi-exponential-kernel benchmark, and (iii) the Gaussian-kernel benchmark.

2 High-demand stress-test validation

The high-demand validation sweep contains all $4 \times 3 \times 3 = 36$ combinations of square command level, release duration, and prepared history. Controller parameters are fixed across the sweep. Over all 36 cases, CF-FOCBF remains hard-safe, policy-safe, and feasible, while also reducing tracking RMSE relative to the history-blind policy-preserving CBF in every case.

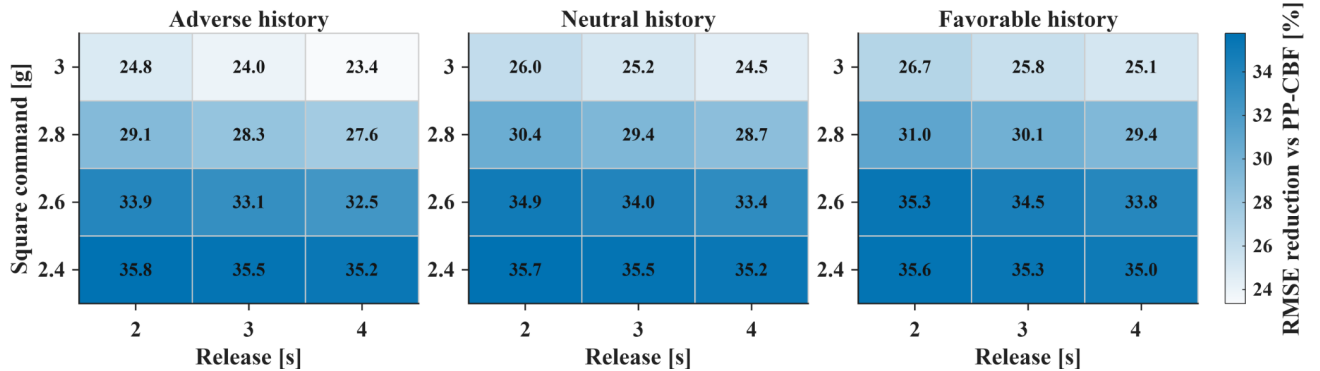


Figure 1: Complete high-demand RMSE-reduction atlas. Each cell reports the percentage reduction of CF-FOCBF relative to the history-blind policy-preserving CBF for the same command, release duration, and prepared history.

3 Beyond CF: kernel definitions

For a generic nonsingular fading-memory kernel, the retained-history coordinate is

$$q_{\kappa}(t) = \int_{t_0}^t \kappa(t - \tau) \dot{h}(\tau) d\tau. \quad (1)$$

The supplementary study compares the following three kernels.

$$\kappa_{\text{CF}}(s) = \kappa_0 e^{-\lambda s}, \quad (2)$$

$$\kappa_{\text{BE}}(s) = \kappa_0 [a e^{-\lambda_1 s} + (1 - a) e^{-\lambda_2 s}], \quad (3)$$

$$\kappa_{\text{G}}(s) = \kappa_0 \exp \left[- \left(\frac{s}{\tau_G} \right)^2 \right]. \quad (4)$$

The CF kernel admits the exact one-state realization

$$\dot{q} = -\lambda q + \kappa_0 \dot{h}. \quad (5)$$

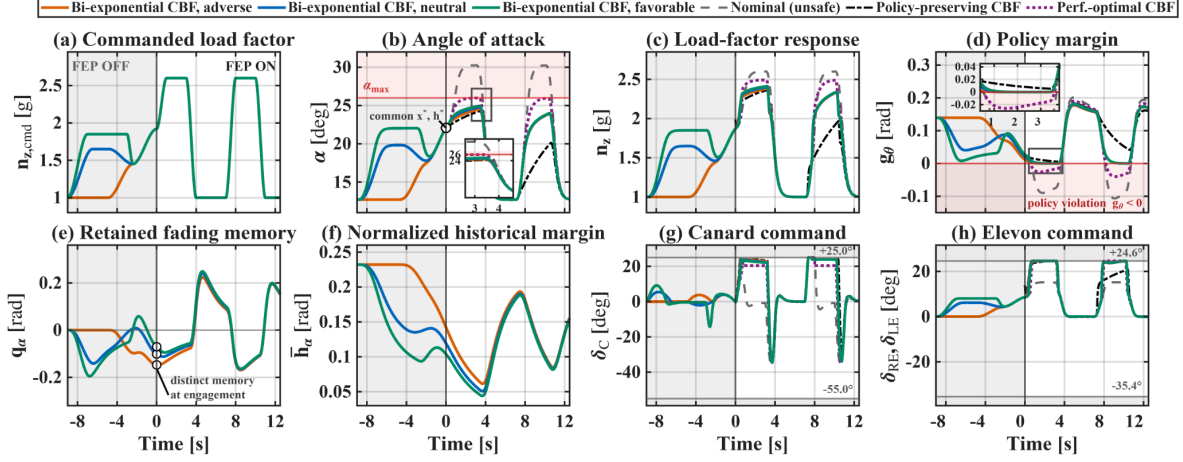
The bi-exponential kernel admits the exact two-state realization

$$\dot{q}_i = -\lambda_i q_i + \kappa_0 \dot{h}, \quad q = a q_1 + (1 - a) q_2. \quad (6)$$

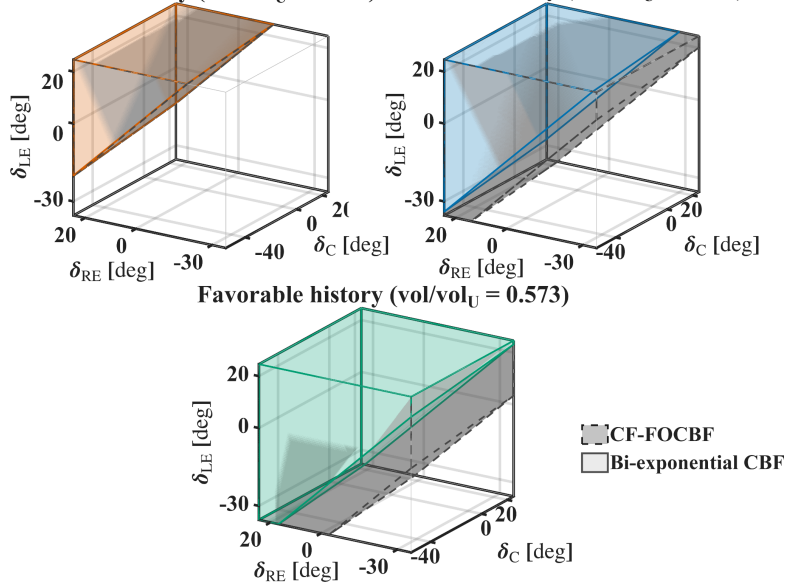
The Gaussian kernel is evaluated directly from retained history through the convolution integral itself, without replacing it by a finite-dimensional exponential surrogate.

4 Bi-exponential kernel benchmark

The same preparation protocol, common activation-state comparison, and twin-square maneuver are reused with the bi-exponential kernel. The adverse < neutral < favorable ordering is preserved both in the retained-history response and in the same-state admissible-control volume.



(a) Twin-square benchmark under the bi-exponential kernel.
Adverse history ($\text{vol}/\text{vol}_U = 0.239$) Neutral history ($\text{vol}/\text{vol}_U = 0.427$)

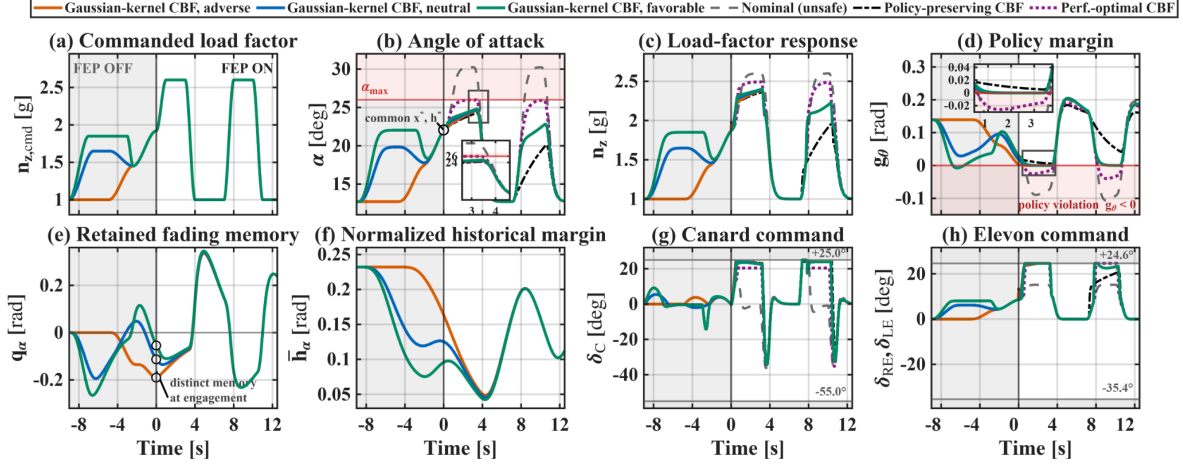


(b) Same-state admissible-set comparison under the bi-exponential kernel.

Figure 2: Bi-exponential-kernel validation. The mechanism remains visible with an exact two-state fading-memory realization.

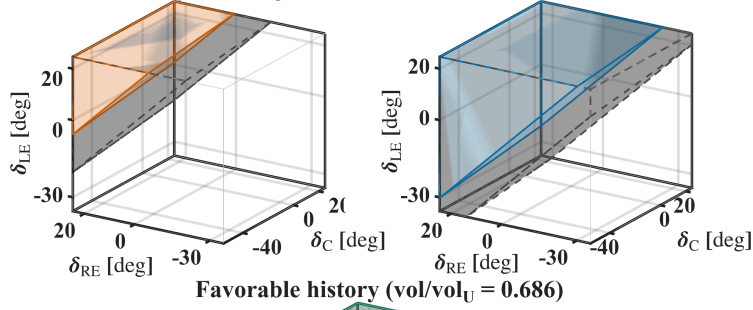
5 Gaussian kernel benchmark

The Gaussian benchmark removes reliance on an exact finite-dimensional exponential realization and evaluates the memory term directly from retained history. The history-conditioned ordering again persists, demonstrating that the effect is not tied to the one-state CF implementation.

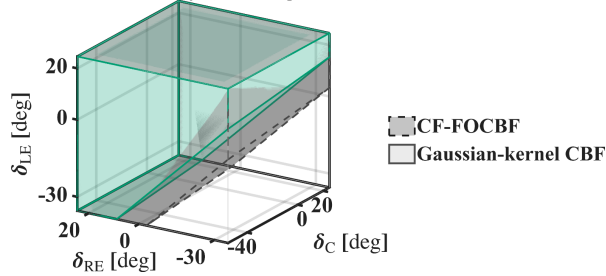


(a) Twin-square benchmark under the Gaussian kernel.

Adverse history ($\text{vol}/\text{vol}_U = 0.101$) Neutral history ($\text{vol}/\text{vol}_U = 0.370$)



Favorable history ($\text{vol}/\text{vol}_U = 0.686$)



(b) Same-state admissible-set comparison under the Gaussian kernel.

Figure 3: Gaussian-kernel validation. The retained-history mechanism remains visible even when the kernel is evaluated directly in the history domain.

6 Availability

The public GitHub Pages site collects the figures used in the manuscript companion. A cleaned and documented MATLAB package is being prepared for public release. Numerical datasets underlying the reported figures and tables are available from the corresponding author upon reasonable request.